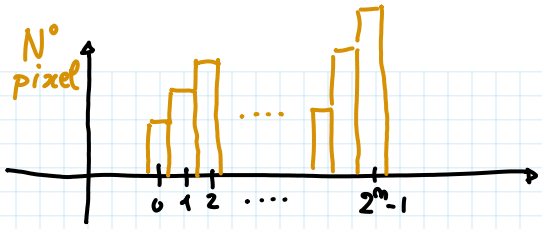
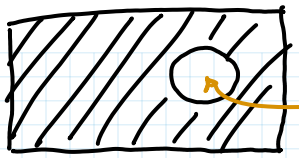
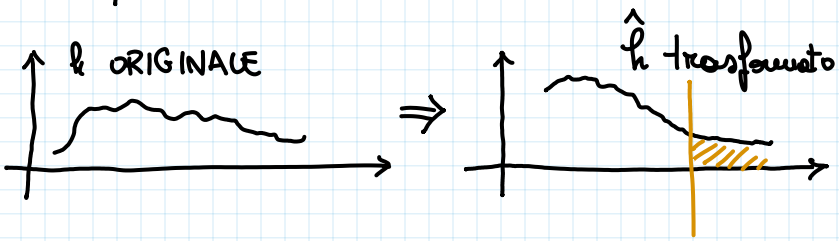




oggetto chiaro

$$t(\ell) = \left| \{ h(\ell) \geq T \} \right|$$

SOGLIA



AREA dell'oggetto

$$\sum_{\ell=0}^{2^m-1} t(\ell)$$

OPERATORI PUNTUALI $\longrightarrow$ TRASFORMAZIONI

$$\{ h_0, h_1, h_2, \dots, h_{2^m-1} \} \overset{\sigma}{\Longrightarrow} \{ \hat{h}_0, \hat{h}_1, \dots, \hat{h}_{2^m-1} \}$$

Def. INVERSIONE dei livelli di grigio

↳ Nel CONTINUO (*) $f: \mathbb{R}^2 \to \mathcal{C}$

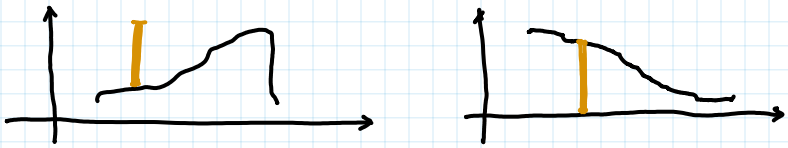
↳ Nel DISCRETO (**) $L \in \mathbb{R}^{H \times W}$

} livelli di grigio $0 \div N_\ell$

(*) Def. una funzione $\boxed{i(t) = N_\ell - t}$ t è l'intensità di un pixel

$$i(f(\underline{x})) = N_\ell - f(\underline{x})$$

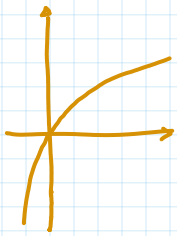
(**) $I \in \mathbb{R}^{H \times W}$ $\boxed{I_{ij} = N_\ell - L_{ij}}$ $i = 1, \dots, H$, $j = 1, \dots, W$ L_{ij} è l'intensità del pixel in posizione (i,j)



Def. COMPRESSIONE LOGARITMICA $\longrightarrow$ ESPANDE i livelli BASSI
COMPRIME i livelli ALTI

↳ Nel CONTINUO:

$$\boxed{\ell(t) = \frac{1}{1+N_\ell} \cdot \log(1+t)}$$



t intensità del pixel

$$\ell(f(\underline{x})) = \frac{1}{1+N_\ell} \log(1 + f(\underline{x}))$$